

Fuel Efficiency Prediction Using Regression-Based Machine Learning: A Case Study on Vehicle Fuel Economy Data

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1) Problem Statement and Background Information

In recent years, rising fuel costs and increasing awareness of environmental impact have pushed the automotive industry to focus heavily on fuel efficiency. Consumers, manufacturers, and policy-makers alike are seeking ways to evaluate and optimize vehicle fuel performance. Accurately predicting fuel efficiency—typically quantified as miles per gallon (MPG)—is therefore a crucial task that has real-world implications across multiple sectors.

Our project addresses the challenge of predicting a vehicle's fuel efficiency using regression-based machine learning models. Specifically, we explore how technical vehicle attributes—such as engine size, horsepower, cylinder count, and fuel type—can be used to accurately predict MPG. Unlike traditional analytical approaches that may rely on simplistic averages or limited rule-based formulas, our work leverages the predictive capabilities of linear regression models enhanced with feature selection methods and modern model evaluation techniques.

The dataset used for this study is obtained from the U.S. government's open data portal, titled Vehicle Fuel Economy. It includes over 40,000 vehicles manufactured between 1984 and 2019 and encompasses 93 features across 26 manufacturers. These features include both categorical and numerical attributes, ranging from vehicle size and engine specs to CO₂ tailpipe emissions and fuel types.

As a reference for best practices in building and interpreting regression models, we followed the methodology outlined in Khalid Jeffal's 2021 article "Linear Regression for Predicting MPG". His work provided guidance on linear regression model construction and evaluation using metrics such as RMSE and R^2 . We adopted his methodology but extended it through more comprehensive preprocessing, feature engineering, and validation strategies to ensure a more robust and generalizable model.

2) Data Loading

The raw dataset was initially provided in CSV format (vehicle-fuel-economy.csv). In line with the course requirements and to demonstrate a complete data pipeline, the data was first loaded into a MySQL database before being imported into the Python environment for analysis and modeling. The following steps were taken to load the required files:

1. A dedicated MySQL database named cap4770 was created.
2. A SQL schema (setup.sql) was written to define the structure of the vehicle_fuel_economy table. This schema included appropriate data types for each of the 93 attributes, balancing text, integers, and floating-point numbers.
3. The data was inserted using the LOAD DATA LOCAL INFILE command, which efficiently bulk-imports CSV files.

In Python, the mysql-connector-python library was used to connect the Jupyter Notebook to the database. The connection was configured using host, user credentials, and database selection. Once connected, the entire dataset was queried using:

```
SELECT * FROM vehicle_fuel_economy;
```

This query was passed to `pandas.read_sql_query()`, which returned the dataset as a `DataFrame`. This method of data import promotes reproducibility and ensures compatibility with future database-oriented data mining pipelines.

By integrating MySQL into the data pipeline, this project mirrors industry-standard ETL workflows (Extract, Transform, Load), where data is first normalized and stored in databases before being passed to analytics or machine learning layers.

3) Data Collection and Preprocessing

Data Collection:

The dataset is derived from the EPA Fuel Economy database, curated by the U.S. Department of Energy and the Environmental Protection Agency. It represents a comprehensive record of fuel efficiency test results and manufacturer-reported data. Key characteristics of the dataset includes:

- 40,705 vehicle records
- 93 attributes such as miles per gallon, make, model, and annual emissions.
- Time span: model years from 1984 to 2019

Brands: includes data from 26 vehicle manufacturers

Data Preprocessing:

The dataset, though rich, contained inconsistencies, irrelevant fields, and missing values typical of real-world data. The preprocessing phase involved the following steps:

Attribute Selection: Attributes unrelated to fuel efficiency (e.g., createdOn, modifiedOn, id) were dropped. Columns with excessive missing or zero values were also removed.

Missing Values and Anomalies: Several attributes such as horsepower and displacement had zero values that do not make physical sense. These rows were either imputed using median values (where feasible) or removed based on domain understanding.

Outlier Detection: MPG values above 100 were flagged as likely outliers caused by electric vehicles or data errors. These cases were filtered or separately analyzed to prevent skewing the regression model.

Normalization and Scaling: For attributes with significantly different ranges (e.g., co2TailpipeGpm vs. horsepower), standardization was applied to ensure balanced feature weighting during model training.

4) Exploring the Data

Our group wanted to understand the underlying structure and relationships in the data. Through exploratory data analysis (EDA), we were able to identify patterns, validate assumptions, and discover relationships between key vehicle attributes and MPG.

Feature analysis

The following list enumerates the twenty six most relevant and consistent attributes of the dataset, It also includes their type, count number, constraint. The features not listed included several null and inconsistent values.

Data columns (total 27 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	Annual fuel consumption (B)	40704 non-null	float64
1	MPG (City)	40704 non-null	int64
2	co2TailpipeGpm	40704 non-null	float64
3	MPG (Combined)	40704 non-null	int64
4	cylinders	40524 non-null	float64
5	displacement	40526 non-null	float64
6	engId	40704 non-null	int64
7	engine description	24667 non-null	object
8	Fuel Efficiency score	40704 non-null	int64
9	fuel annnual cost08	40704 non-null	int64
10	fuel annual costA	40704 non-null	int64
11	fuelType	40704 non-null	object
12	fuelType1	40704 non-null	object
13	MPG (highway)	40704 non-null	int64
14	hybrid level	40704 non-null	int64
15	horsepower	40704 non-null	int64
16	id	40704 non-null	int64
17	make	40704 non-null	object
18	model	40704 non-null	object
19	transmission	40693 non-null	object
20	UCity	40704 non-null	float64
21	UHighway	40704 non-null	float64
22	VClass	40704 non-null	object
23	year	40704 non-null	int64
24	youSaveSpend	40704 non-null	int64
25	createdOn	40704 non-null	object
26	modifiedOn	40704 non-null	object

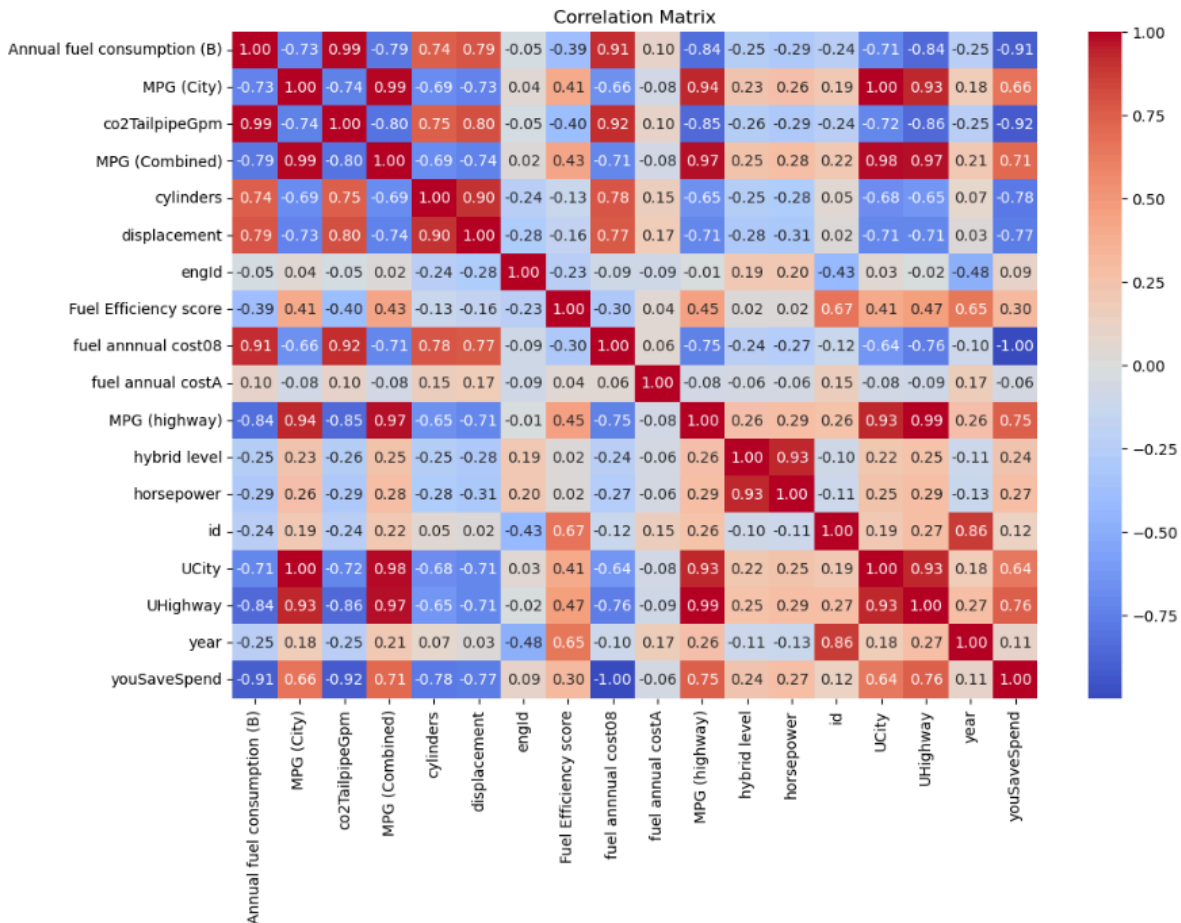
Statistical Summaries:

Descriptive statistics provided initial insights into the range, central tendency, and variance of attributes such as MPG_Combined, engine_displacement, and cylinders.

	MPG (Combined)	Annual fuel consumption (B)	co2TailpipeGpm	displacement	cylinders
count	40704.000000	40704.000000	40704.000000	40526.000000	40524.000000
mean	20.525550	17.325764	467.504446	3.299097	5.720610
std	7.369281	4.602734	123.032961	1.359130	1.755344
min	7.000000	0.060000	0.000000	0.000000	2.000000
25%	17.000000	14.330870	386.391304	2.200000	4.000000
50%	20.000000	16.480500	444.350000	3.000000	6.000000
75%	23.000000	19.388824	522.764706	4.300000	6.000000
max	136.000000	47.087143	1269.571429	8.400000	16.000000

Correlation Analysis:

We calculated the Pearson correlation matrix to identify which variables had the strongest linear relationship with the target variable (MPG Combined). A heatmap visualization revealed that horsepower, cylinders, and displacement had a negative correlation with MPG.



Time Series Trends: A line plot of average MPG_Combined over the years (1984–2019) showed an upward trend, especially post-2005, likely due to stricter fuel economy regulations and advancements in hybrid/electric vehicle technology.

Categorical Analysis:

After comparing MPG across different fuel types we discovered that electric and hybrid vehicles dramatically outperform traditional gas-powered ones in terms of fuel efficiency. As a result, some outlier removal or segmentation (e.g., focusing only on regular and premium gasoline) was necessary for consistent modeling.

5)Data Modeling

The modeling phase focused on predicting the vehicle's combined MPG using multiple regression techniques. We approached this through both baseline linear regression and regularized regression (LassoCV), with added cross-validation to ensure model reliability.

A. Multiple Linear Regression

Multiple linear regression is a model used to estimate the value of a selected variable (dependent variable) by analyzing its correlation with two or more independent variables. The model illustrated in this report aims to predict the value of miles per gallon (combined) by creating a model that analyzes its relationship with five highly correlated variables.

B. Recursive Feature Elimination

Recursive Feature Elimination is a selection method that ranks each feature of the dataset based on their relevance to the dependent variable. After ranking the features it recursively removes the least relevant features until the desired number of features is reached. The model illustrated in this report implements the Recursive Feature Elimination included in the sklearn library and trains the model with selected data. The result of the calculation is the selection of five highly correlated features: Annual fuel consumption, MPG(City), engine displacement, fuel efficiency score, and MPG (highway).

C. Lasso Regression

According to IBM Lasso Regression “facilitates automatic feature selection by adding a penalty term to the residual sum of squares (RSS), which is then multiplied by the regularization parameter (lambda or λ). This regularization parameter controls the amount of regularization applied. Larger values of lambda increase the penalty, shrinking more of the coefficients towards zero; this subsequently reduces the importance of (or altogether eliminates) some of the features

from the model, resulting in automatic feature selection. Conversely, smaller values of lambda reduce the effect of the penalty, retaining more features within the model.” In the model illustrated in the report Lasso Regression is implemented using a cross validation number of 5 and the model is train with the selected train subset. The result of the model implementation is the selection of two features: youSaveSpend with a value of 0.00035 and co2Tailpipegpm with a value of -0.030082

6) Results and interpretation

A. Evaluation Metrics

- Mean Squared Error (**MSE**) is an evaluation metric that calculates the squared difference between the predicted values and the testing values. The result of the calculation could be a number between 0 to infinity. A low value indicates a small difference between the predicted value and the test value.
- Mean Absolute Error (**MAE**) is an evaluation metric that measures the average magnitude of the errors between predicted values and actual values. When the Mean Absolute Error is calculated it does not take into account if the predicted value was lower or higher than the actual value. The result of the calculation could be a number between 0 and infinity. The closer the number is to zero the more accurate the model is.
- R-Squared (R^2) is a statistical measure used to determine the proportion of variance in a dependent variable that can be predicted or explained by an independent variable. In other words, R-Squared shows how well a regression model (independent variable) predicts the outcome of observed data (dependent variable).(Abba, 2023)

Results

	Recursive Feature Elimination	Lasso Regression
Mean Squared Error	0.13107	4.03898
Mean Absolute Error	0.30064	1.27823
R-Squared	0.99511	0.84918

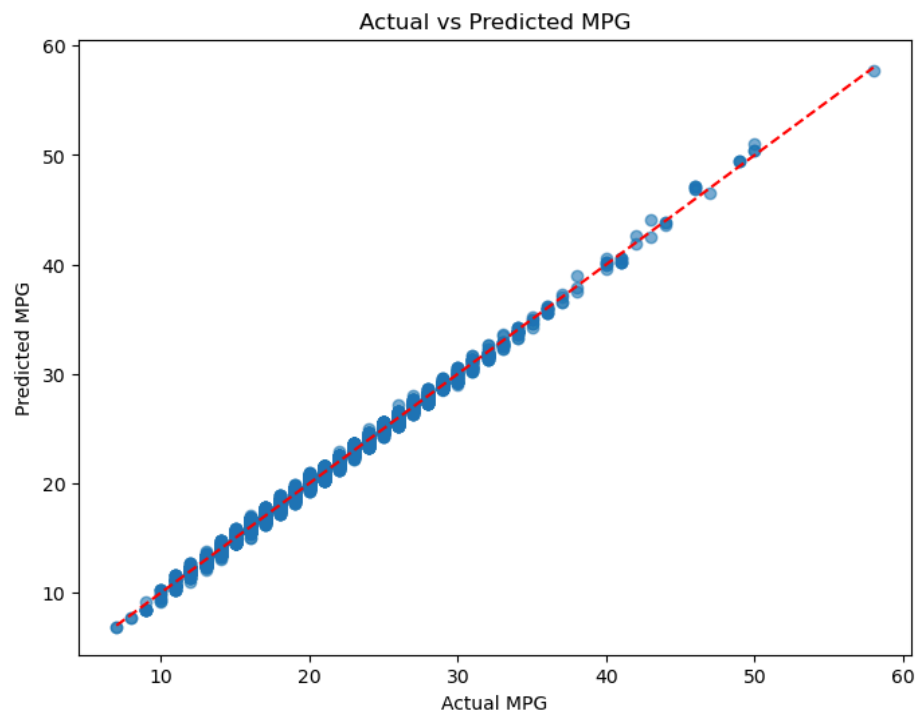
B. Cross Validation

Cross-validation is a statistical method used to evaluate the performance of machine learning models by dividing the dataset into multiple subsets. Instead of training and testing on a single split, cross-validation trains the model multiple times, each time using a different part of the data as the test set and the remaining as the training set. The proposed model divided the dataset into five different subsets and calculated the average Mean Squared Error, Mean Absolute Error, and R-Squared score of each subset. The table below displays the result obtain in the three regression models.

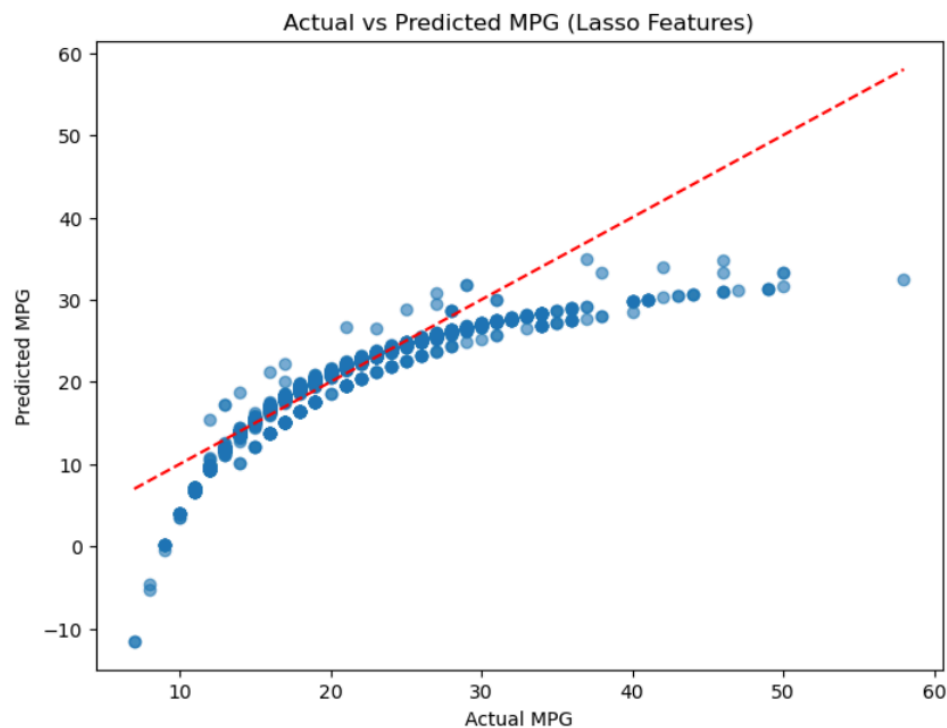
Cross validation results

	Recursive Feature Elimination	Lasso Regression
Mean Squared Error	0.15383	8.13081
Mean Absolute Error	0.32110	1.82279
R-Squared	0.98374	0.15924

C. Recursive Feature Elimination Model



D. Lasso Regression Model



7)References and Citation:

IBM (January, 2024) What is lasso regression, *IBM*. retrieved from:

<https://www.ibm.com/think/topics/lasso-regression#:~:text=Lasso%20regression%20is%20a%20regularization,regularization%20for%20linear%20regression%20models.>

Abba I. (March, 2023) What is r squared? r2 value meaning and definition, *freecodecamp*.

Retrieved from :

<https://www.freecodecamp.org/news/what-is-r-squared-r2-value-meaning-and-definition/>